

GNIOT Institute of Management Studies – PGDM (Batch 2023-25)**(TRIMESTER –IV) END TERM EXAMINATION, OCTOBER -2024****Business Intelligence and Data mining****[Time: 2 Hours]****[Maximum Marks: 60]****Following are the course outcomes of the subject: - Business Intelligence and Data mining**

CO1	Understanding the fundamentals of Business Intelligence empowers businesses to leverage data for better decision-making
CO2	Articulate data warehousing that can support decision-making processes in a business.
CO3	Understanding data mining allows businesses to extract valuable insights from large datasets for data-driven decision making.
CO4	Interpret classification and prediction methods to make accurate predictions, leading to better decision-making in various industries.

(Note- Faculty can add the more CO's as per their subject.)**SECTION-A****Attempt all parts of the following: (Short Answer Type Questions, do not exceed 150- 200 words)**

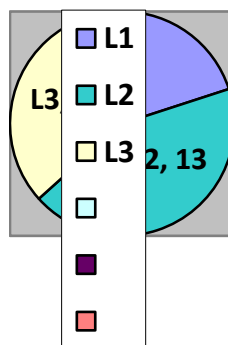
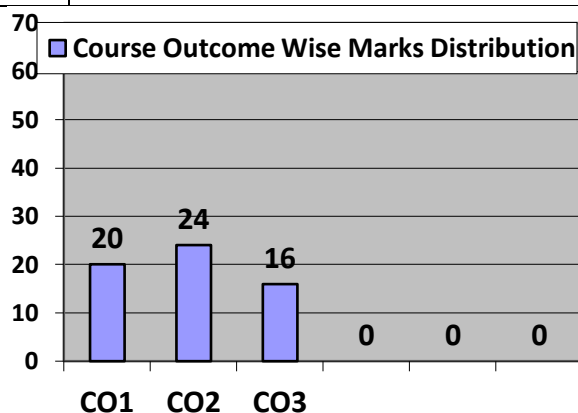
Q.1	Marks	COURSE OUTCOME	BLOOMS LEVEL
(A)	4	CO - 1	BL-1
(B)	4	CO - 1	BL-1
(C)	4	CO - 2	BL-1
(D)	4	CO - 3	BL-1
(E)	4	CO - 4	BL-1

SECTION-B**Attempt ANY TWO questions out of the following: (Long Answer Type Questions, do not exceed 300- 400 words)**

Q. No	Marks	COURSE OUTCOME	BLOOMS LEVEL
2 (A)	10	CO – 4	BL-3
2 (B)	10	CO – 4	BL-3
3 (A)	10	CO – 4	BL-3
3 (B)	10	CO – 4	BL-3

SECTION-C**Attempt all parts of the following: (Caselet / Numerical, do not exceed 300- 400 words)**

Q.4	Marks	COURSE OUTCOME	BLOOMS LEVEL
4 (A)	10	CO – 2	BL-3
4 (B)	10	CO - 2	BL-3

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GNIOT Institute of Management Studies – PGDM (Batch 2023-25)**(TRIMESTER – IV) END TERM EXAMINATION, OCTOBER - 2024****Business Intelligence and Data Mining****[Time: 2 Hours]****[Maximum Marks: 60]****INSTRUCTIONS:**

- (i) **There are THREE sections in this question paper.**
- (ii) **Attempt questions from each section as per the directions.**
- (iii) **Make suitable assumptions wherever necessary.**

SECTION-A**(Short Answer Type Questions, do not exceed 200 words)****Q.1. Attempt all parts of the following:****(5×4=20)**

- (a) Define Business Intelligence (BI).
- (b) Describe the key information sources that existed before the emergence of BI.
- (c) Explain the evolution of Business Intelligence.
- (d) Outline how BI architecture has developed over time.
- (a) Summarize the concepts and processes involved in Data Warehousing.

SECTION-B**(Long Answer Type Questions)****Q.2. Attempt all questions of the following****(2X10=20)**

RetailX is a retail company specializing in consumer electronics and household goods. The company operates both physical stores and an online platform. RetailX has accumulated transaction data over the past year and wants to perform a descriptive analysis to understand customer purchase behavior, identify trends, and make data-driven decisions to enhance its sales and marketing strategies.

Sample data:

Transaction ID	CustomerID	ItemID	Quantity	Price	TransactionDate
1	101	A001	2	50.00	2024-01-05
1	101	B002	1	30.00	2024-01-05
2	102	A001	1	50.00	2024-01-10
3	103	C003	3	20.00	2024-01-15
3	103	B002	2	30.00	2024-01-15
4	104	A001	1	50.00	2024-01-20
4	104	C003	2	20.00	2024-01-20

- (a) What type of plot would you use to visualize sales distribution by item?

OR

- (b) How do you generate this plot using seaborn or matplotlib?

Q.3. (a) Describe how you would create a line plot to show monthly transaction counts.**OR**

- (b) What insights can this plot provide about sales trends?

SECTION-C

GlobalTech Inc. is a multinational manufacturing company specializing in electronics. The company operates multiple production facilities and warehouses around the world, and its supply chain involves complex logistics, including procurement, production, inventory management, and distribution.

GlobalTech Inc. has been facing challenges with supply chain inefficiencies, including high inventory costs, delays in production, and distribution issues. To address these challenges and improve overall efficiency, the company decides to implement **OLAP (Online Analytical Processing)** to analyze and optimize its supply chain operations.

Q.4. Attempt all parts of the following:

(2×10=20)

To utilize OLAP to gain insights into GlobalTech's supply chain operations and identify key areas for improvement. The goals are to:

4(a) Analyze procurement and inventory data to reduce carrying costs.

4(b) Identify bottlenecks in the production process.